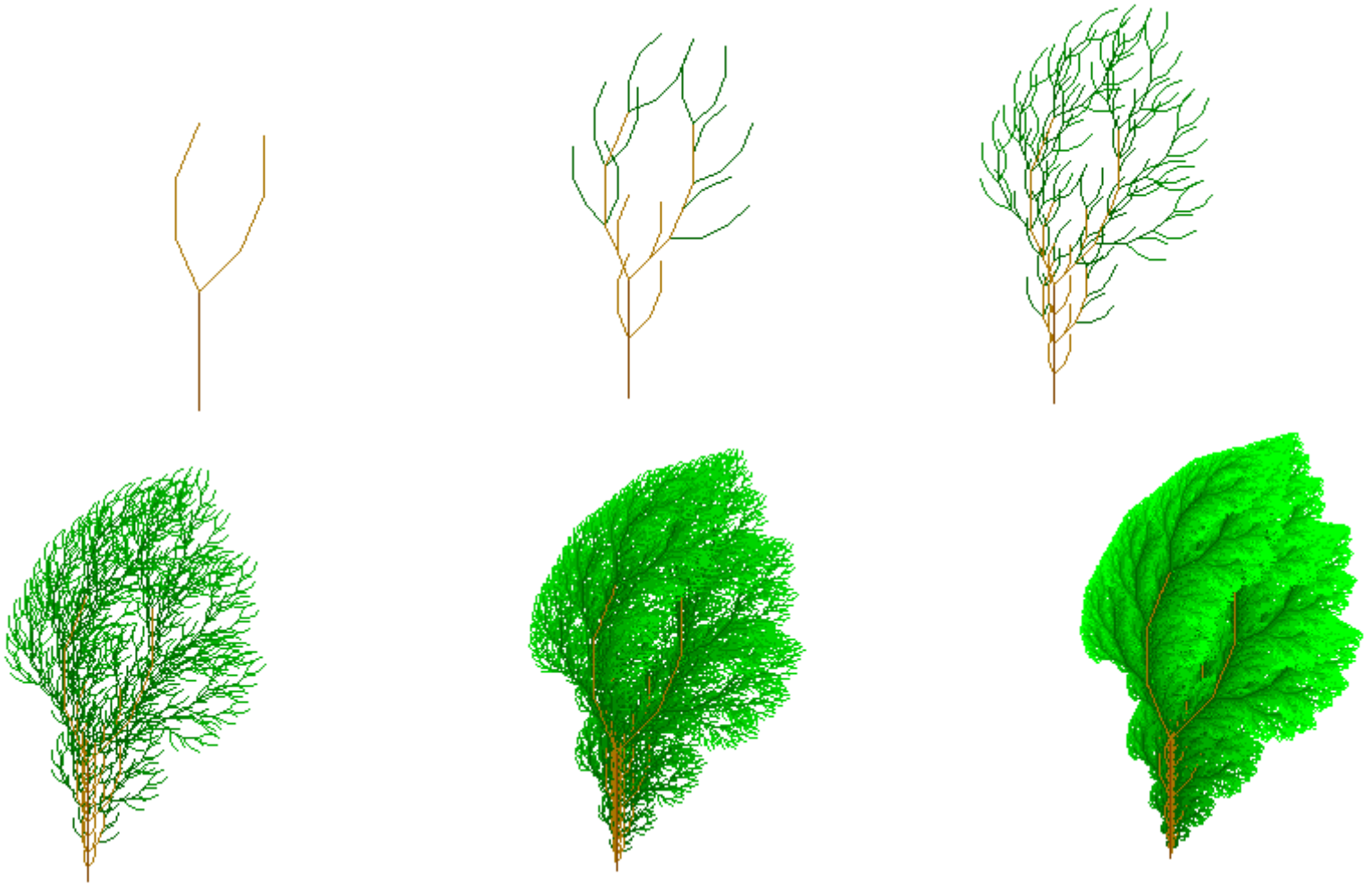
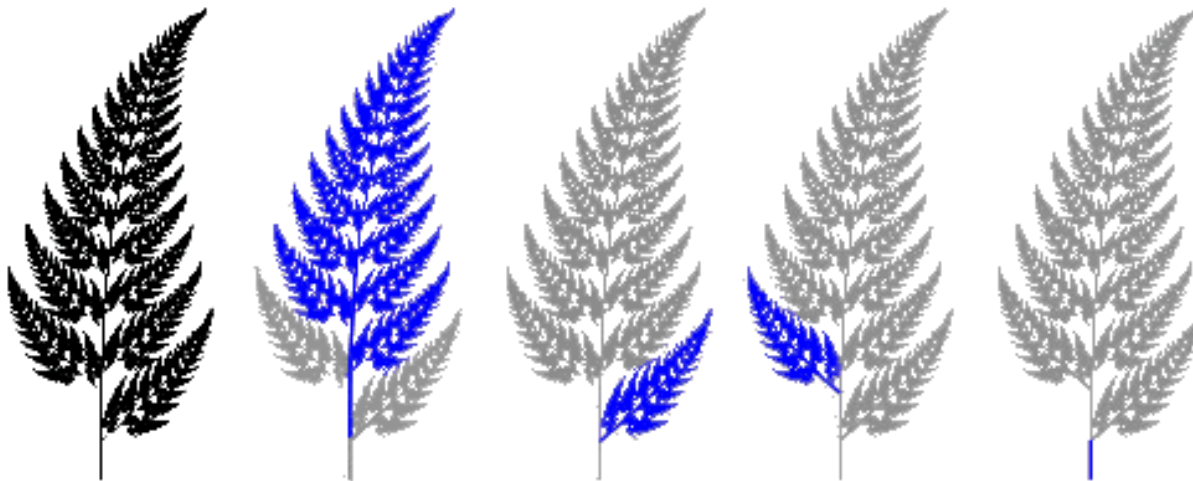
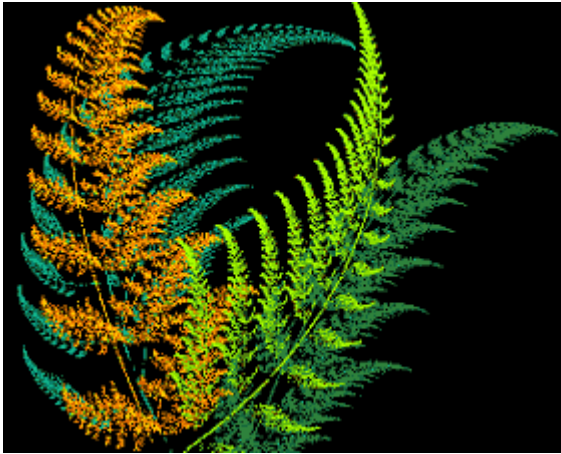


Representation and Modeling of 3D Objects (P2)

Fractals

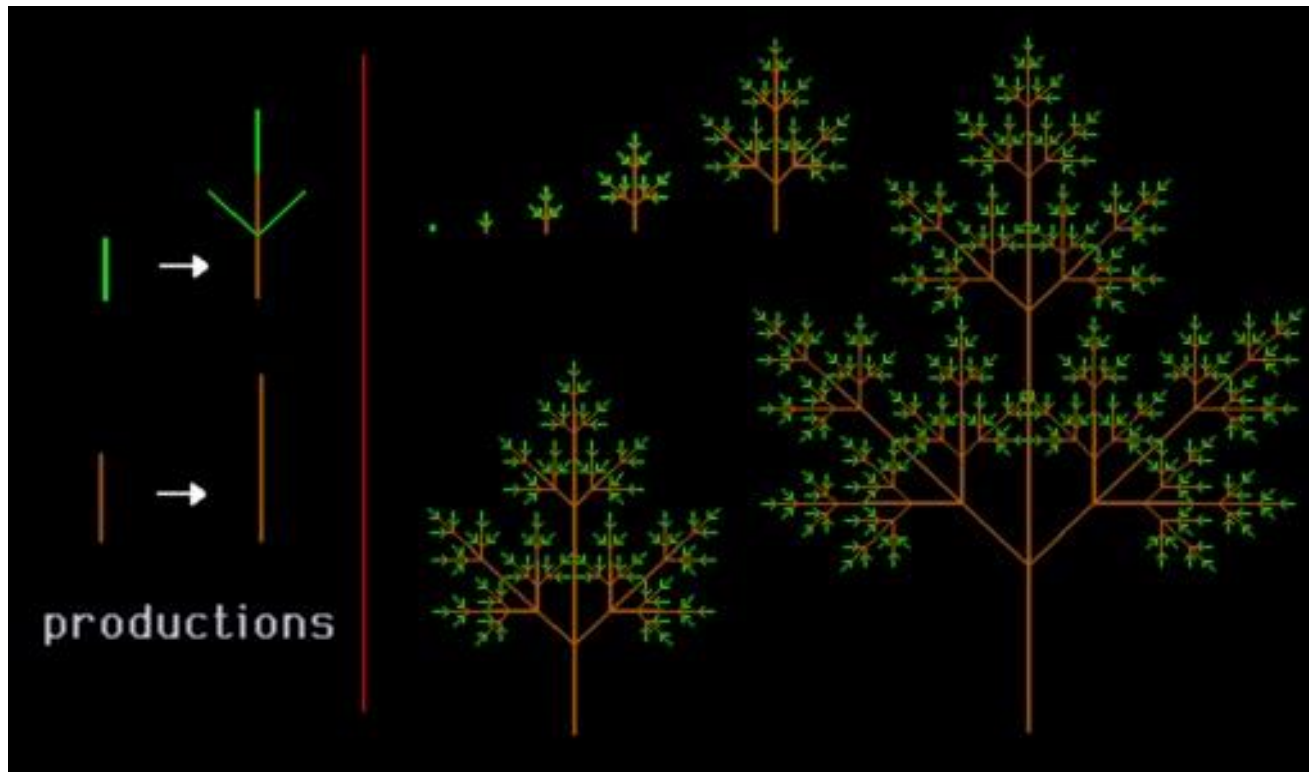


Fractals



Procedural modeling

- ❑ Lindenmayer-System or L-System
- ❑ Prusinkiewicz, P., Lindenmayer, A., Hanan, J.: "Development Models of Herbaceous Plants for Computer Imagery Purposes", SIGGRAPH 88, pp. 141-150
- ❑ Examples from Prusinkiewicz, Hammel, Mech. <http://www.cpsc.ucalgary.ca/projects>

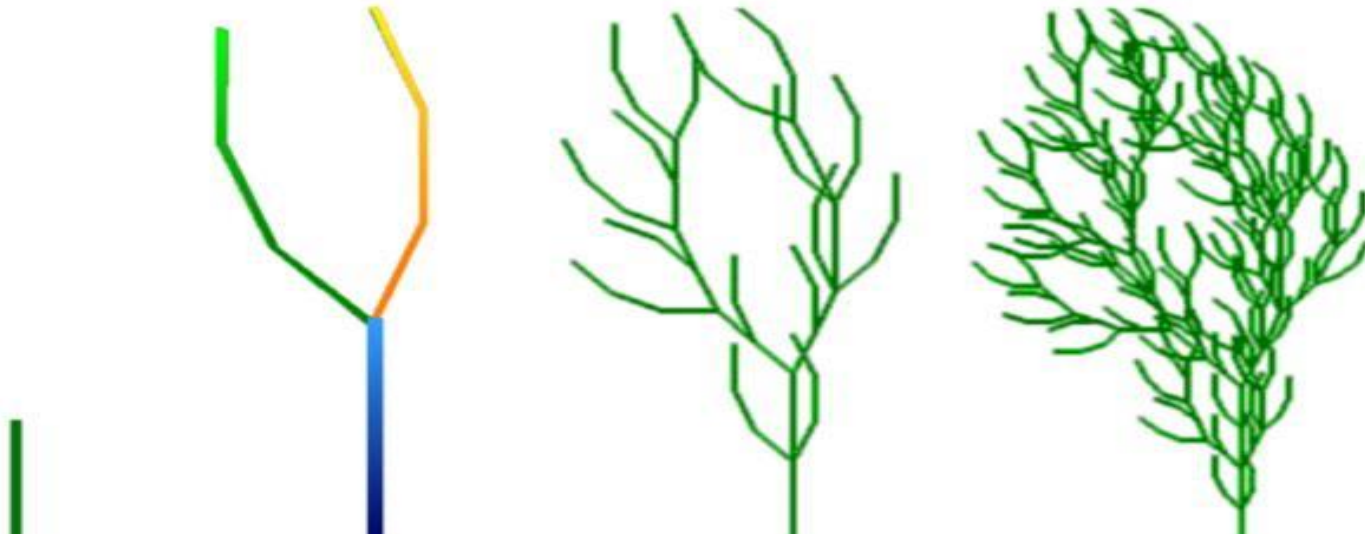


Lindenmayer-System or L-System

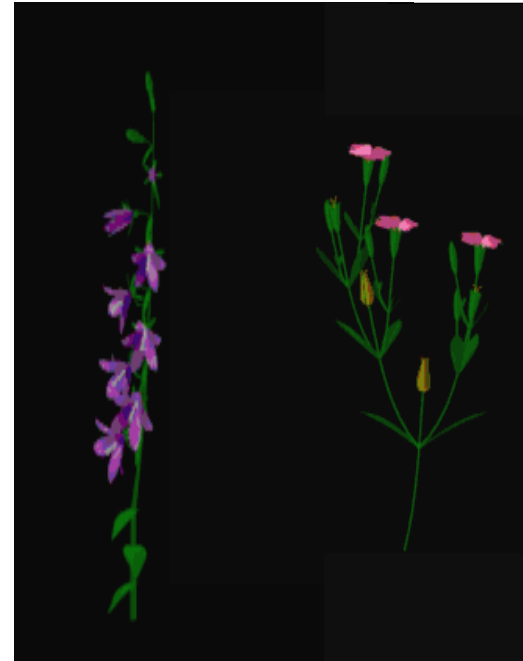
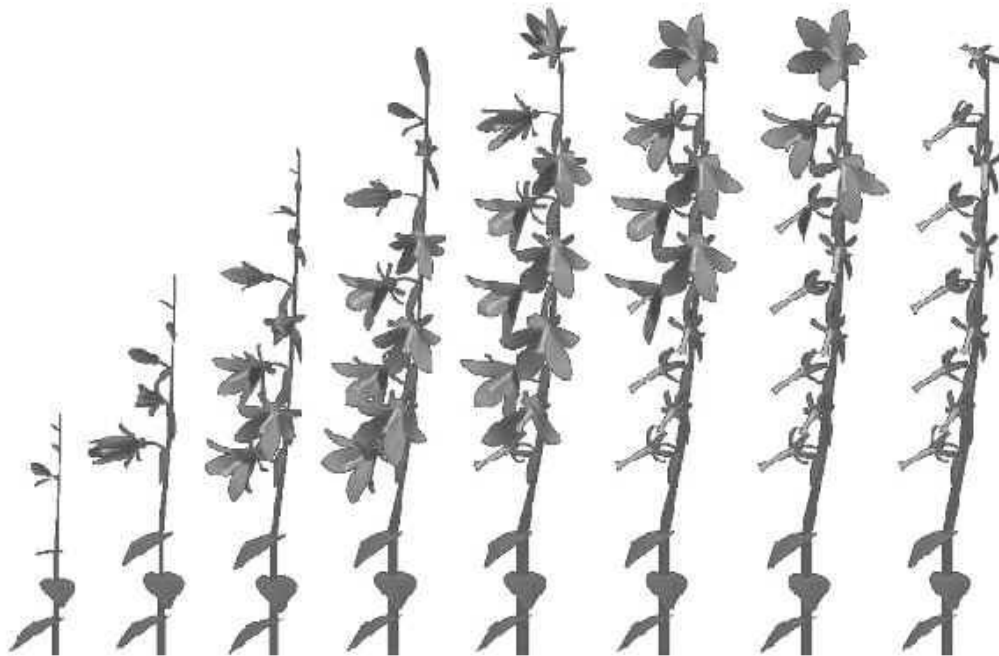
□ Production rules:

W -> **F**

F -> **FF**+[**+F-F-F**] -[**-F+F+F**]



Lindenmayer-System or L-System



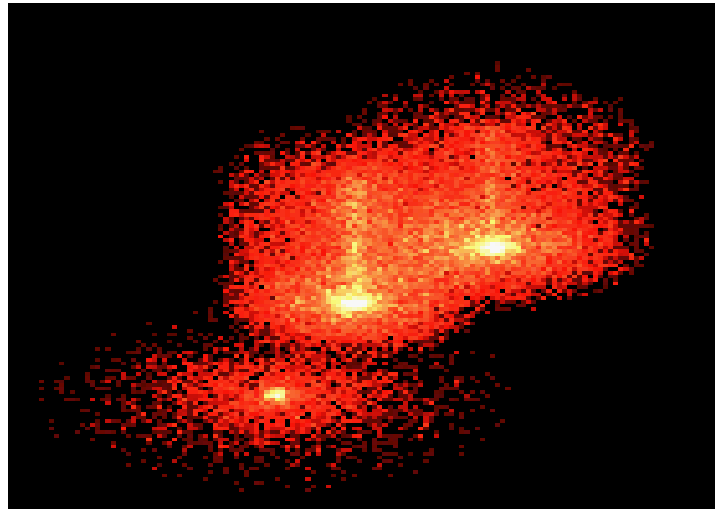
Lindenmayer-System or L-System



Procedural modeling

□ Particle based systems

- The term particle system is loosely defined in computer graphics. It has been used to describe modeling techniques, rendering techniques, and even types of animation. In fact, the definition of a particle system seems to depend on the application that it is being used for.
- Describe natural or irregular shaped objects which are flowing or fluid-like, particular objects which are changing over time
- Model objects like grass, waterfalls, fireworks, cloth, etc.

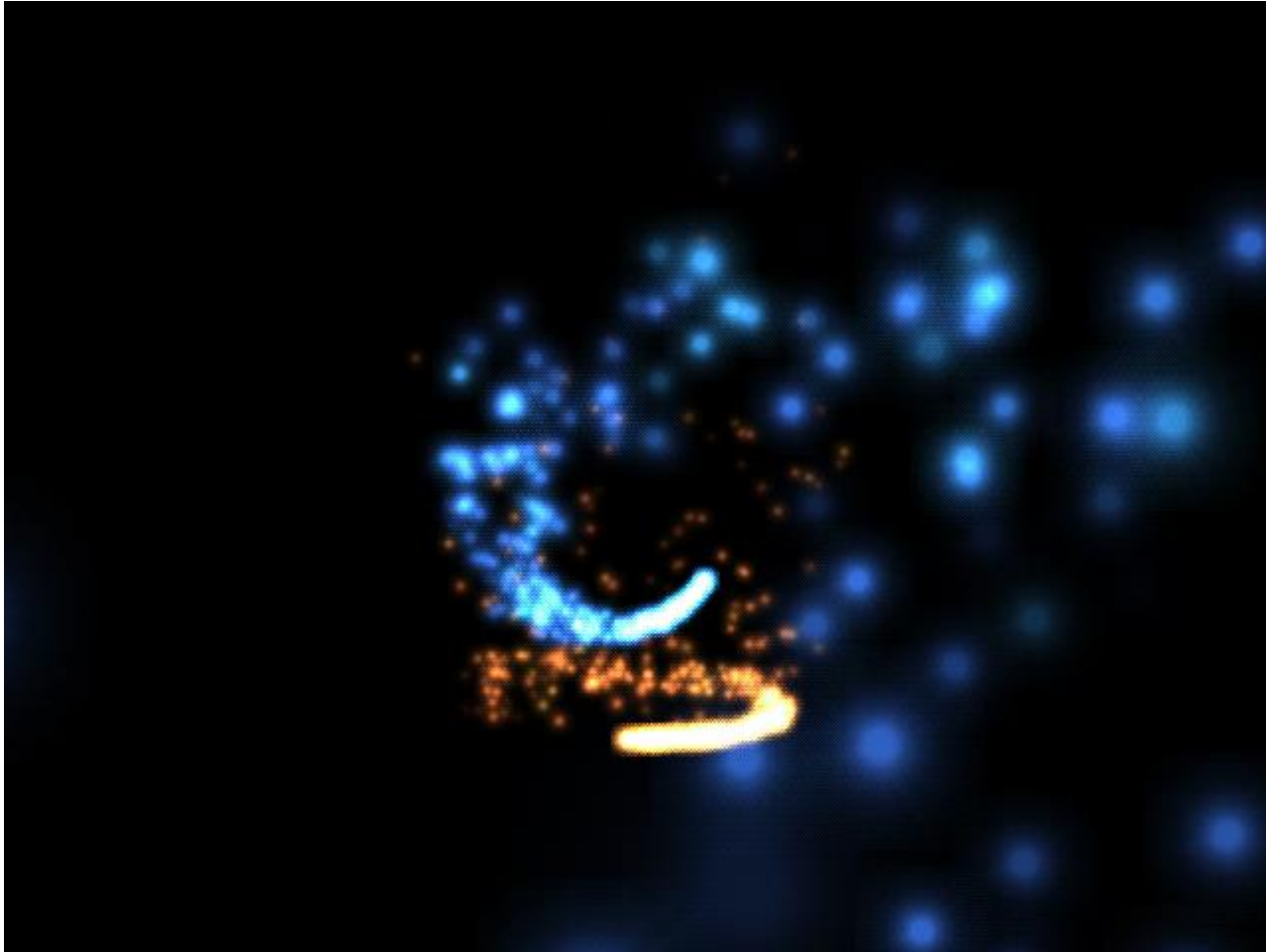


Particle based systems

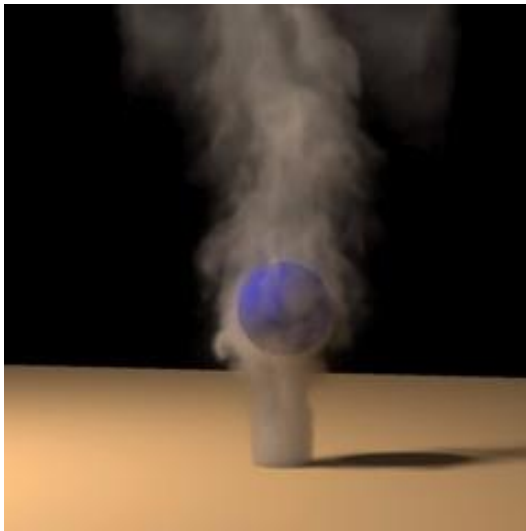
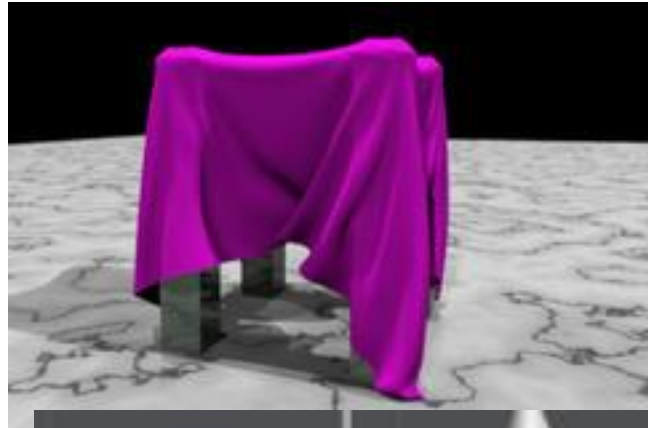
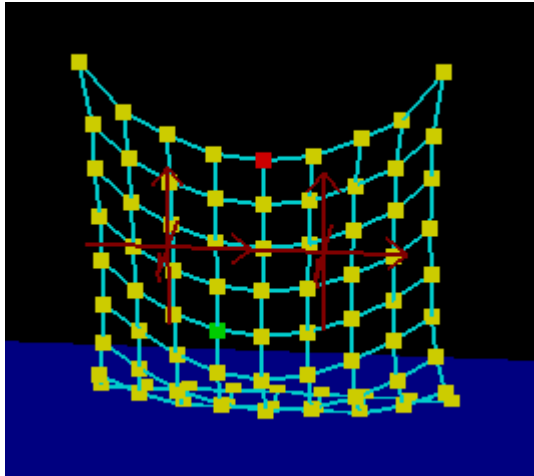
- ❑ Collection of particles - a particle system is composed of one or more individual particles.
- ❑ Stochastically defined attributes : position, velocity (speed and direction), color, lifetime, age, shape, size, transparency
- ❑ Particle Life Cycle
 1. Generation
 - particles in the system are generated randomly within a predetermined location of the fuzzy object
 2. Particle Dynamics
 - the attributes of each of the particles may vary over time. (particle position is going to be dependent on previous particle position and velocity as well as time)
 3. Extinction
 - each particle has two attributes dealing with length of existence: age and lifetime.
 4. Rendering
 - when rendering this system of thousands of particles, some assumptions have to be made to simplify the process.

Particle based modeling

ENERGiE - Particle Engine v2.26. Bua Manuel, <http://www.Planet-Source-Code.com>

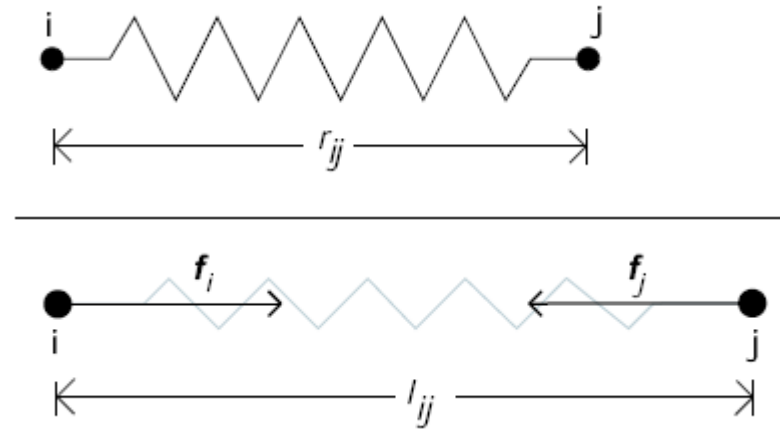


Particle based modeling

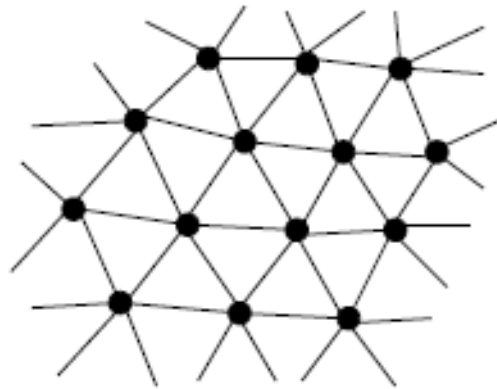


Mass-spring particle model

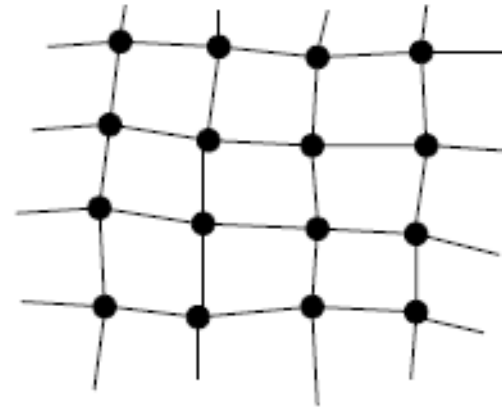
- Connected particle network
- The link between two nodes **i** and **j** is an arc
- $\Delta x = k x$



Mass-spring particle model



a)



b)

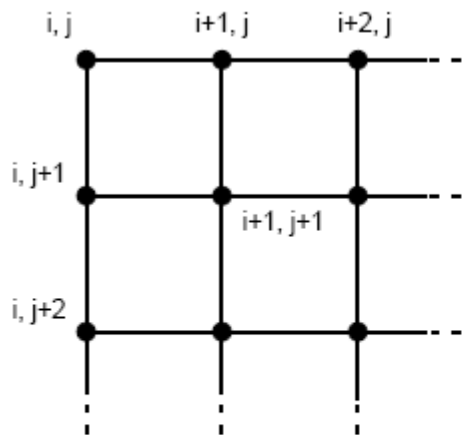
Particle network a) triangular topology b) rectangular topology.

Simulation:

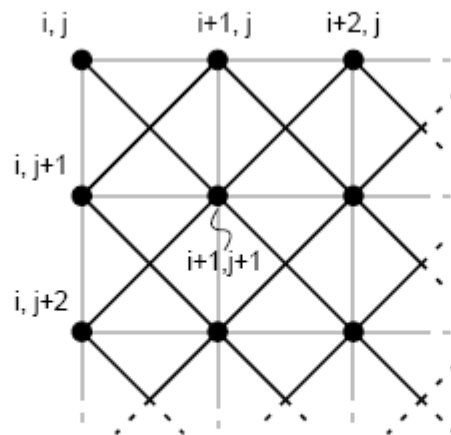
1. Force computation
2. Collision detection
3. Graphics rendering

Mass-spring particle model

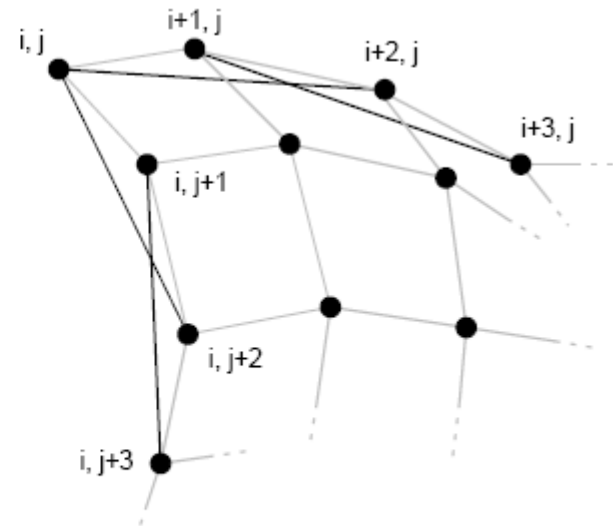
- Links between nodes (arcs):
 - Structural/Tensile springs (Strutturale)
 - Shear springs (Intindere)
 - Bend springs (Indoire)



a) Structural/Tensile springs



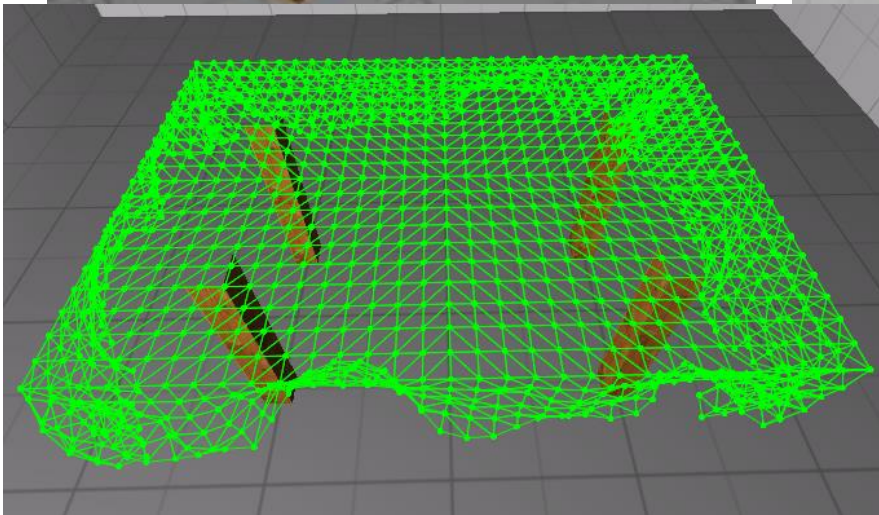
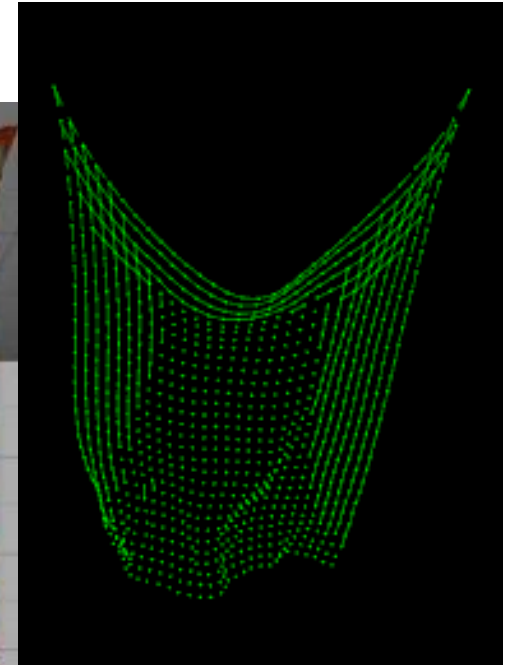
b) Shear springs



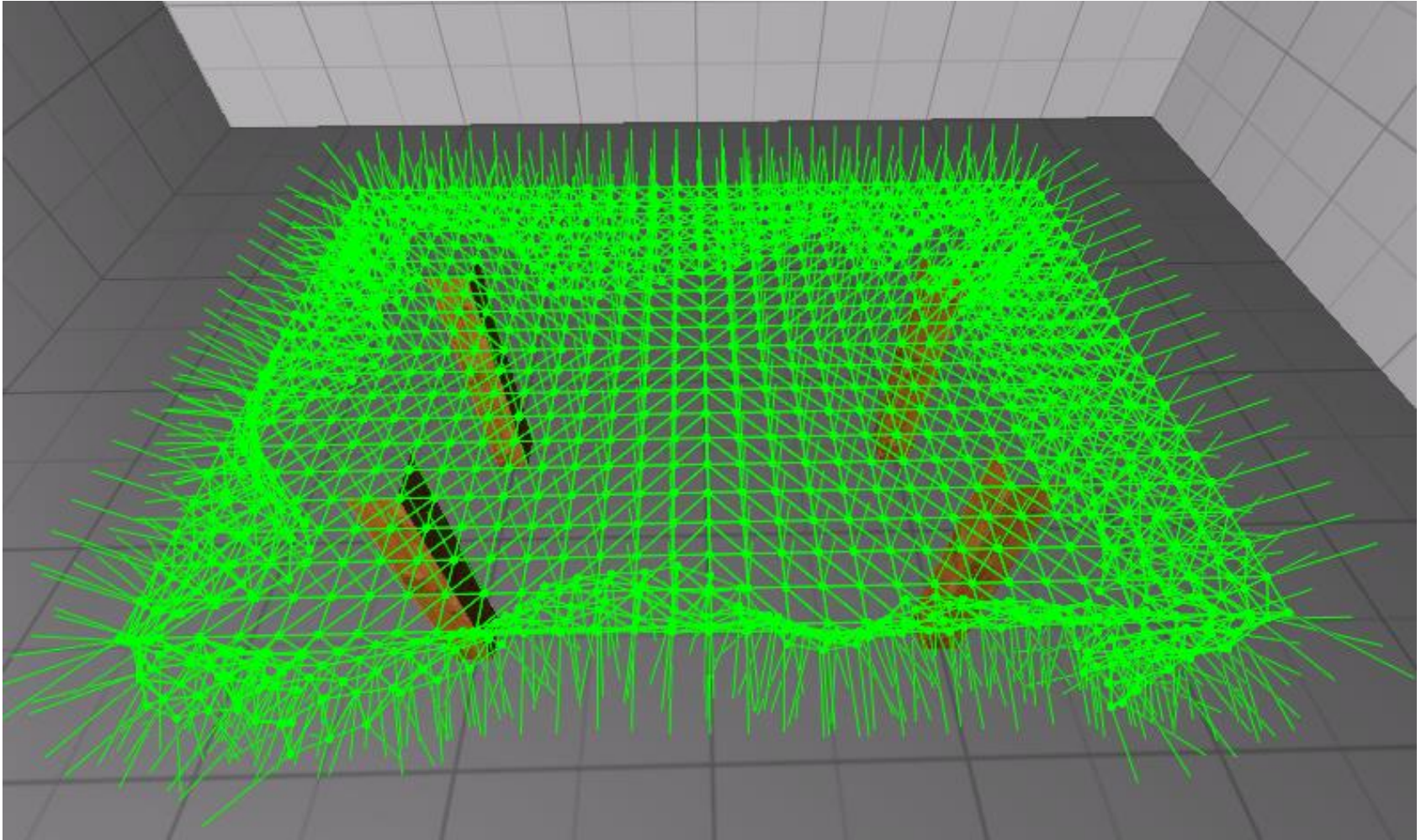
c) Bend springs
(not all connections shown)

Particle based modeling

Cloth simulation: ClothSim, <http://www.benve.org/ClothSim>



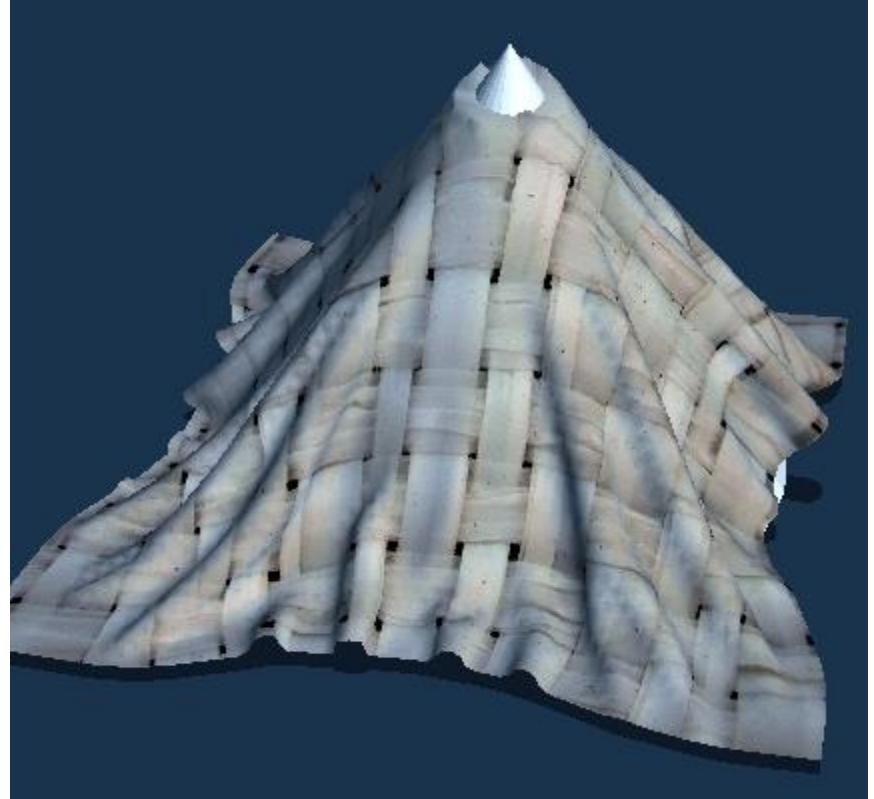
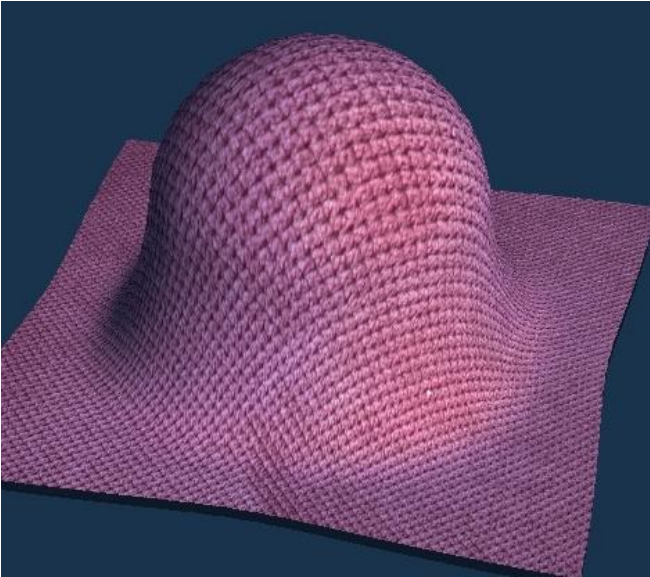
Particle based modeling - ClothSim



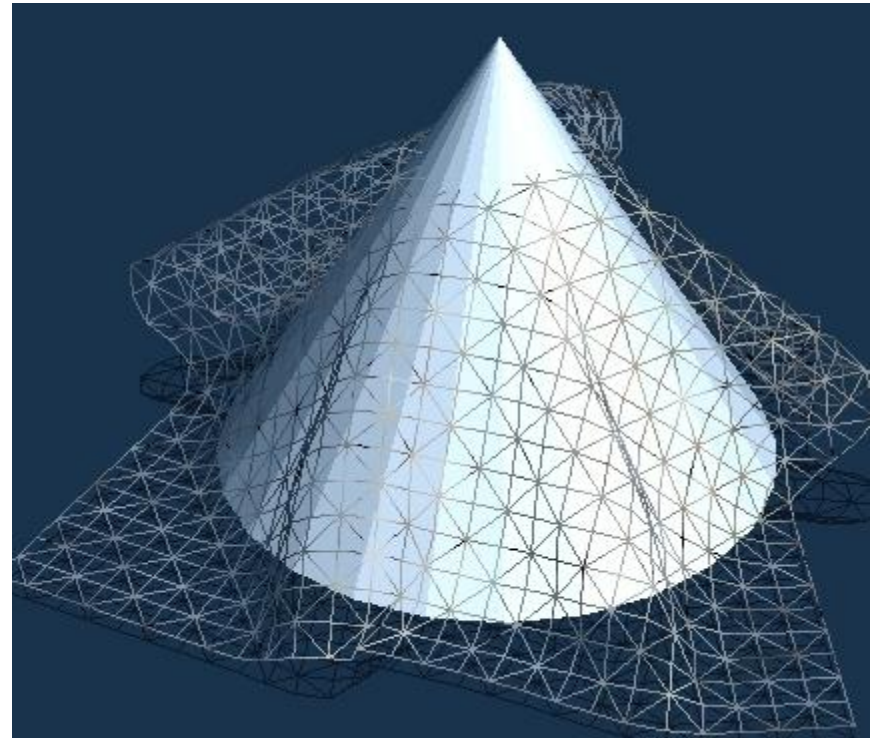
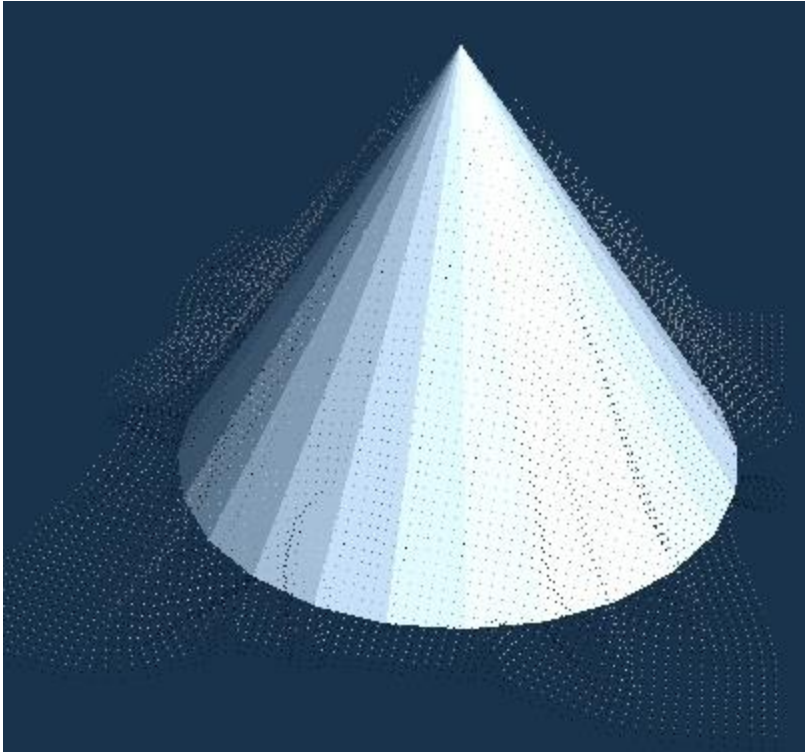
Particle based modeling



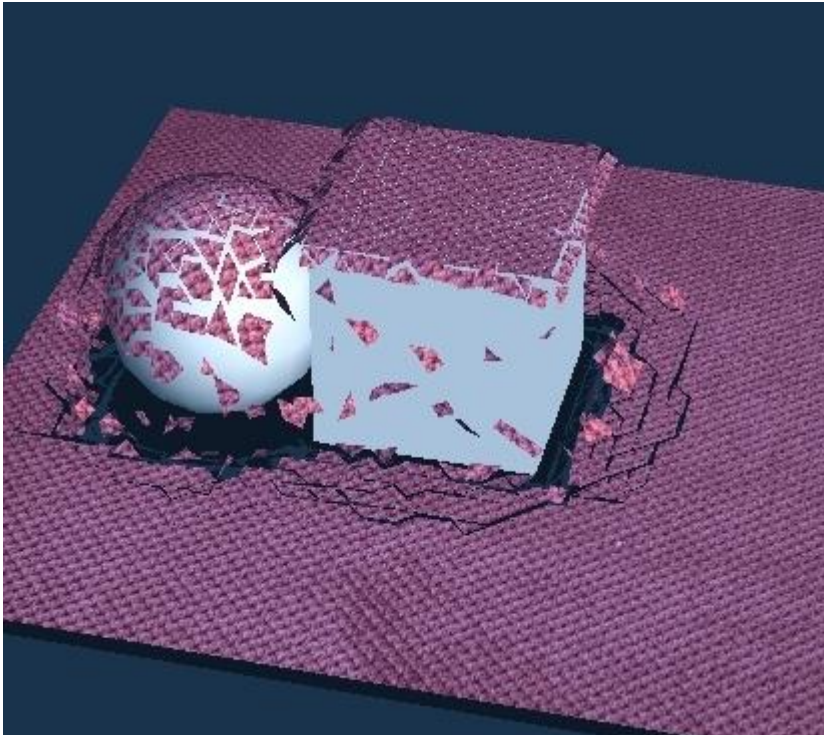
Particle based modeling



Particle based modeling



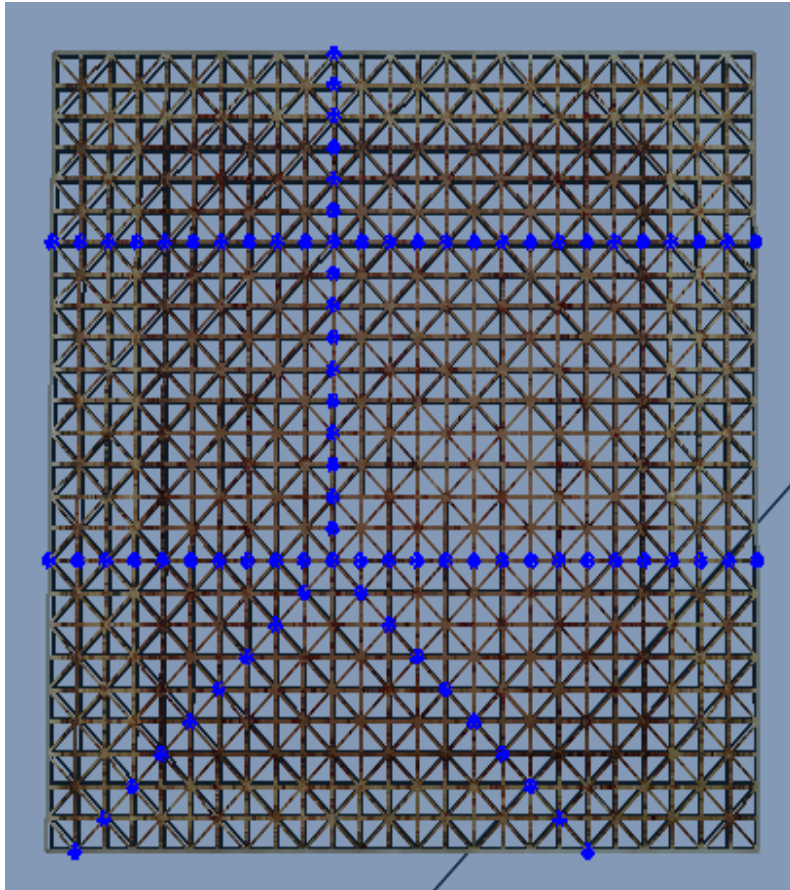
Particle based modeling



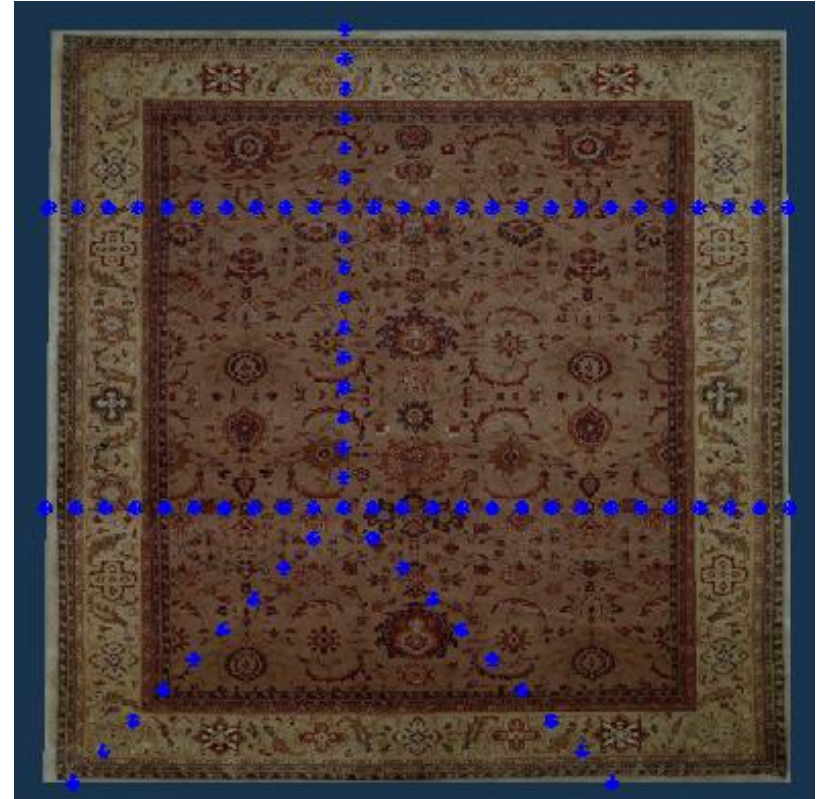
Particle based modeling



Particle based modeling - cutting

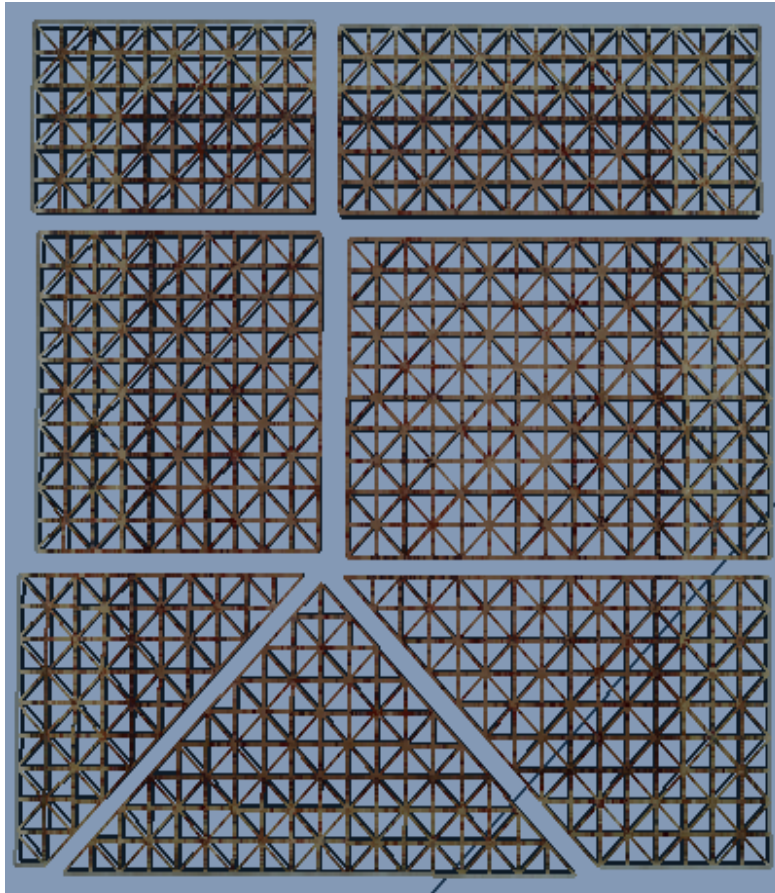


Selection of the cutting lines. Wireframe visualization.



Selection of the cutting lines. Texture mapping visualization.

Particle based modeling - cutting



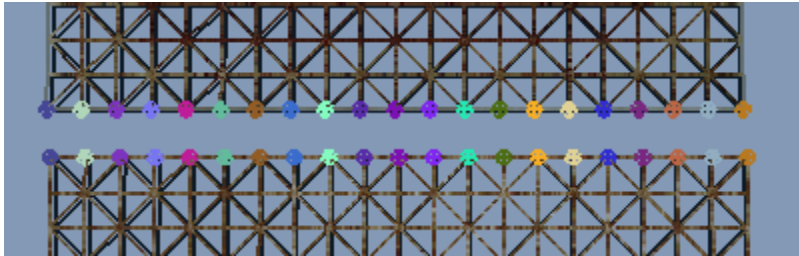
Cutting operation. *Wireframe visualization.*



Cutting operation. *Texture mapping visualization.*

Particle based modeling - sewing

Equal steps:



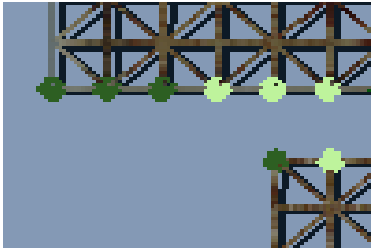
Preview of the cutting operation.
Wireframe visualization.



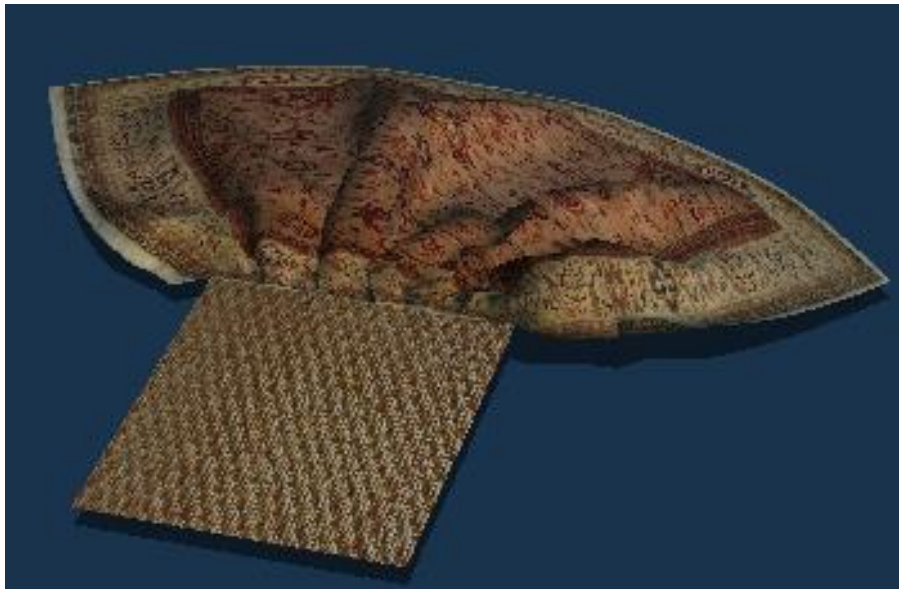
Sewing operation. Constant sewing
step, equal one.

Particle based modeling - sewing

Sewing step equal three.



Preview of the cutting operation. Constant sewing step, equal three.



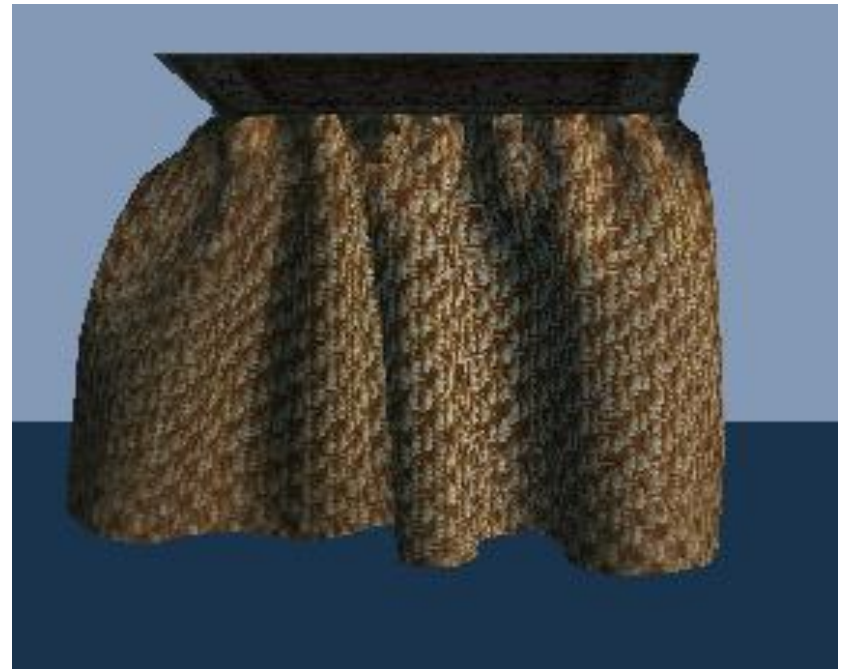
Sewing operation. Constant sewing step, equal three (Goga, 2009).

Particle based modeling - sewing

Variable sewing step.



Preview of the sewing operation. Variable sewing step, equal three and respectively two (Goga, 2009).



Sewing operation. Variable sewing step, equal three and respectively two (Goga, 2009).

Cloth on animated 3D objects



Baraff and Witkin, "Walking man and Dancer with long skirt", SIGGRAPH 98 Conference.